



1-Butanol synthesis from ethanol over strontium phosphate hydroxyapatite catalysts with various Sr/P ratios

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ABSTRACT

Strontium phosphate hydroxyapatites (Sr-HAP) with various Sr/P molar ratios were synthesized under hydrothermal conditions. X-ray photoelectron spectroscopy revealed a close correlation between the near-surface and bulk Sr/P molar ratios of catalyst samples. Sr-HAP catalysts show high 1-butanol selectivity in the gas-phase conversion of ethanol. Sr-HAP catalysts with higher Sr/P molar ratios showed higher catalytic activity and 1-butanol selectivity. The density of both relatively strong basic sites and acidic sites was seen to increase with increasing Sr/P molar ratios, though the basic site density was significantly higher than the acidic site density. The rate-determining step in ethanol conversion over Sr-HAP catalysts was considered to be the dimerization process, as aldol condensation is mainly accelerated by base catalysis, which would explain why the Sr-HAP catalysts with higher basic site density showed higher catalytic activity and 1-butanol selectivity.

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1. Introduction

Calcium phosphate hydroxyapatite (Ca-HAP) has a high thermal stability and an affinity for organic compounds. It is used widely in fields such as biomaterials, adsorbents, and chromatography. The stoichiometric form of Ca-HAP is shown as $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$, and its Ca/P molar ratio is 1.67. However, the ionic radius of the component elements allows a fair degree of transfer or loss of ions within its crystal structure, making Ca-HAP highly nonstoichiometric, with a Ca/P molar ratio ranging from 1.5 to 1.7 [1]. It was postulated that calcium-deficient Ca-HAP ($\text{Ca/P} < 1.67$) is facilitated by the fact that the loss of Ca^{2+} ions and the resulting electrical imbalance are corrected by introduction of other cations, such as H^+ , and by changing OH^- groups to H_2O groups, and this is represented by the formula $\text{Ca}_{10-n}(\text{HPO}_4)_n(\text{PO}_4)_{6-n}(\text{OH})_{2-n}(\text{H}_2\text{O})_n$ [1,2]. On the other hand, calcium-rich Ca-HAP ($\text{Ca/P} > 1.67$) is thought to be a mixture of stoichiometric Ca-HAP and $\text{Ca}(\text{OH})_2$, or a solid solution resulting from the partial replacement of PO_4^{3-} ions with other anions, such as CO_3^{2-} ions [1,3], although evidential data of this for these compounds have not been detected by powder X-ray diffraction and infrared spectroscopy measurements. Some researchers have reported that Ca-HAP shows properties of acid–base catalysis and that the catalysis depends on its Ca/P molar ratio [2,4,5]. Ca-HAP catalysts with low Ca/P molar ratios show

acidic catalysis, whereas Ca-HAP catalysts with high Ca/P molar ratios show basic catalysis. Moreover, Ca-HAP is reported to have a high capacity for replacement of Ca^{2+} ions and PO_4^{3-} ions by other cations and anions, respectively [6–10], and its acid–base catalysis is changed by substitution of Sr^{2+} , Ba^{2+} , Pb^{2+} , Cd^{2+} , and VO_4^{3-} ions [10,11]. For example, strontium phosphate hydroxyapatite (Sr-HAP) catalyst shows significantly higher acetone selectivity than that over Ca-HAP catalyst in the catalytic conversion of 2-propanol, which suggests that Sr-HAP catalyst has significantly more basic catalysis [10].

Biomass-derived ethanol is expected to become a major renewable feedstock. Its conversion into valuable chemicals is expected to replace fossil resources. Recently, the selective conversion of ethanol into 1-butanol over Ca-HAP catalysts was reported [12–14]. In this catalytic system, it was reported that the catalytic activity and selectivity of Ca-HAP catalysts are affected by the Ca/P molar ratios of the catalysts, and Ca-HAP catalysts with higher Ca/P molar ratios exhibited high catalytic activity and 1-butanol selectivity [14]. On the other hand, we examined the catalytic properties of various substituted hydroxyapatite catalysts for catalytic conversion of ethanol, and we found that the Sr-HAP catalyst showed higher 1-butanol selectivity than Ca-HAP catalyst [15]. We therefore expected that a highly active and selective catalyst for 1-butanol formation could be developed by controlling the Sr/P molar ratio of Sr-HAP catalyst.

It is postulated that the reaction mechanism of 1-butanol formation from ethanol includes the following three processes

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(Scheme 1) [14–20]: dehydrogenation of ethanol into acetaldehyde, aldol condensation of acetaldehyde into crotonaldehyde, and then hydrogenation of crotonaldehyde into 1-butanol. In liquid-phase reactions of ethanol using a Raney Ni catalyst with potassium hydroxide, it was reported that at relatively low temperatures (130–140 °C), the ethanol dehydrogenation process is the rate-determining step, while at higher temperatures (160–180 °C), the aldol condensation process is the rate-determining step [21]. In liquid-phase reactions of ethanol using a copper catalyst with sodium methoxide at 180–220 °C, the addition of N₂ to the reaction atmosphere containing H₂ resulted in better catalytic performance, indirectly confirming that the dehydrogenation of alcohols to the corresponding aldehydes is the rate-determining step [22]. However, in gas-phase reactions at around 300 °C, the rate-determining step of 1-butanol formation from ethanol has not been confirmed.

In the present work, we clarified the effect of the Sr/P molar ratio on the catalytic properties of Sr-HAP and the rate-determining step of 1-butanol formation from ethanol in gas-phase reactions.

2. Experimental

2.1. Catalyst preparation

All starting materials were reagent grade chemicals (Wako Pure Chem., Ltd., Co.), such as NaOH (>97.0%), P₂O₅ (>98.0%) and Sr(NO₃)₂ (>98.0%). Strontium phosphate hydroxyapatite (Sr-HAP) with various Sr/P molar ratios was prepared using hydrothermal methods described previously [10,15]. First, a solution containing strontium nitrate (3–4 mmol) in 8 ml of distilled water was added into a solution containing P₂O₅ (1 mmol) in 7 ml of distilled water with NaOH (7 mmol), such that for each preparation the Sr/P molar ratio in the mixed solution was between 1.5 and 2.0. Sr-HAP-NH₃ catalysts were prepared by the same method as Sr-HAP catalysts except for using NH₃ instead of NaOH as the alkaline source. The resultant suspensions were treated under hydrothermal conditions at 110 °C for 14 h with agitation in an autoclave lined with Teflon. After hydrothermal treatment, the resultant precipitates were each washed by centrifugal separation with distilled water and then dried at 60 °C for 5 h.

2.2. Characterization

Unless otherwise noted, the as-prepared powders were characterized by the following techniques. The crystalline phase was identified by powder X-ray diffraction (XRD; Rigaku, Ultima IV) with Cu K α radiation (40 kV and 20 mA). The bulk elemental composition of each sample was determined by inductively coupled plasma (ICP; Shimadzu, ICPE-9000). Approximately 4 mg of sample was dissolved in 20 ml of 1 M HNO₃ solution and then diluted to 100 ml with distilled water. Surface analysis was carried out with X-ray photoelectron spectroscopy (XPS; JEOL, JPS-9010MC) using Al K α radiation. For XPS analysis, samples were pretreated at 550 °C for 3 h in Ar flow and then cooled at room temperature and pelletized to 2 mm ϕ . Because the surface was electrically charged, the binding energies were corrected using 531.1 eV for O1s as an internal standard [23–25]. Infrared (IR) spectra were performed with a JASCO FT/IR 4100 spectrometer in the range

400–4000 cm^{−1} using a KBr disk technique. Specific surface area was measured by nitrogen physisorption using the Brunauer–Emmett–Teller (BET) method (BELSORP-max, BEL Japan, Inc.).

The FT-IR spectrum of adsorbed pyridine species on the sample was recorded on a Thermo Fisher Scientific NICOLET 6700 FT-IR spectrometer. The sample was pretreated at 400 °C for 3 h under vacuum. After dehydration, a background spectrum was recorded, and then, a small volume of pyridine vapor was admitted and allowed to equilibrate. The FT-IR spectrum was recorded after evacuated at 150 °C for 3 h. All spectra were recorded after the sample was cooled to room temperature.

The densities of acidic sites and basic sites were estimated from the amount of strongly adsorbed NH₃ and CO₂, respectively, on samples at 250 °C [14], after pretreatment at 430 °C for 3 h under vacuum. The amounts of NH₃ and CO₂ adsorbed on the samples were determined using a BELSORP-max instrument. Relatively strong acidic site and basic site densities were calculated as in the following equation:

Relatively strongly acidic site or basic site density ($\mu\text{mol m}^{-2}$) = [Amount of NH₃ or CO₂ adsorbed onto fresh sample at 250 °C ($\mu\text{mol g}^{-1}$) – Amount of NH₃ or CO₂ adsorbed onto sample evacuated at 250 °C after the first adsorption ($\mu\text{mol g}^{-1}$)] / Specific surface area of sample ($\text{m}^2 \text{g}^{-1}$).

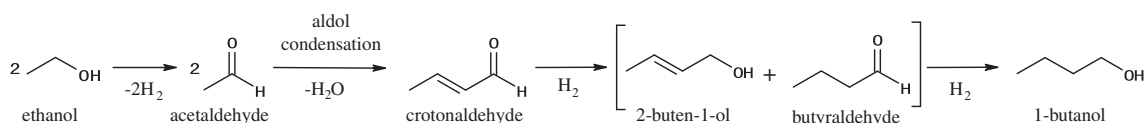
Temperature-programmed desorption (TPD) of CO₂ was carried out using a glass U-tube reactor equipped with a quadrupole mass spectrometer (ANELVA, M-201QA-TDM). Samples (0.1 g) were preheated in flowing He at 550 °C and evacuated and then exposed to 50 kPa of 10% CO₂/He gas at room temperature until saturation coverage was reached. After the sample was flushed with He at room temperature for 30 min, the temperature was then increased at a linear rate of 5 °C/min from 20 to 800 °C under vacuum. The TPD spectra were normalized based on the specific surface area. The TPD of NH₃ was carried out by almost the same method as the TPD of CO₂, except for using 10% NH₃/He gas.

2.3. Catalytic reactions

Catalytic conversion of ethanol was carried out at 300 °C using 2.0 g of catalyst in a fixed-bed continuous-flow glass reactor (8 mm i.d.) under atmospheric pressure. The catalyst powders were pelletized and crushed to the desired size (250–500 μm). Before the catalytic reaction, the as-prepared catalysts were pretreated at 550 °C for 3 h in Ar flow. Ethanol was introduced into the reactor using a micro-syringe pump. The initial concentration of ethanol in Ar was 10.3–53.5 mol%. W/F_{ethanol} was adjusted between 32 and 389 h g_{catalyst} mol^{−1}_{ethanol} by controlling the ethanol flow rate. The typical W/F_{ethanol}, catalyst weight, ethanol flow, and Ar flow were 130 h g_{catalyst} mol^{−1}_{ethanol}, 2.0 g, 15.4 mmol h^{−1}, and 1.8 L h^{−1}. The reaction products were condensed in an ice-water trap and collected for analysis on a GC-FID (Shimadzu GC14B) with a DB-WAX capillary column (60 m, 0.25 mm i.d.). The ethanol conversion, product yield, product selectivities, W/F_{ethanol}, and specific reaction rate were calculated based on the following equations, which were also used in previous reports [12–15]:

Ethanol conversion (C-%)

$$= (1 - \text{C mol of unreacted ethanol} / \text{C mol of the liquid and gas products and unreacted ethanol}) \times 100$$



Scheme 1. The reaction mechanism of 1-butanol formation from ethanol [14–20].

Product yield (C-%)

$$= (\text{C mol of product} / \text{C mol of the liquid and gas products and unreacted ethanol}) \times 100$$

Product selectivity (C-%) = (product yield/ethanol conversion) $\times 100$

W/F_{ethanol} ($\text{h g}_{\text{catalyst}} \text{mol}_{\text{ethanol}}^{-1}$)

$$= \text{catalyst weight(g)} / \text{ethanol flow rate}(\text{mol h}^{-1})$$

specific reaction rate ($\text{mol h}^{-1} \text{m}^{-2}$)

$$= (\text{ethanol conversion}/100) / (\text{specific surface area of catalyst} \times W/F_{\text{ethanol}})$$

The mass balance was calculated as the ratio of the weight of collected liquid to the weight of introduced ethanol. The mass balance in ethanol conversion over the Sr-HAP catalyst was $100 \pm 1\%$.

Catalytic conversion of 1-butanol was carried out at 300°C . The initial concentration of 1-butanol in Ar was 16.1 mol%. The $W/F_{1\text{-butanol}}$, catalyst weight, 1-butanol flow, and Ar flow were $130 \text{ h g}_{\text{catalyst}} \text{mol}_{1\text{-butanol}}^{-1}$, 1.276 g, 9.84 mmol h^{-1} , and 1.14 L h^{-1} . The 1-butanol conversion, product yield, product selectivities, $W/F_{1\text{-butanol}}$, and specific reaction rate were calculated based on the equations for the catalytic conversion of ethanol.

Catalytic conversion of acetaldehyde and crotonaldehyde with 2-butanol was carried out at 300°C using 1.0 g of catalysts. The mixture of substrates, comprising acetaldehyde and crotonaldehyde, and 2-butanol was introduced into the reactor using a micro-syringe pump. The initial concentration of acetaldehyde in Ar was 0.2 mol%, that of crotonaldehyde was 0.1 mol% and that of 2-butanol was 10.8 mol%. The conversion, product yield, and product selectivities were calculated based on the following equations:

Conversion (C-%)

$$= (1 - \text{C mol of unreacted substrate} / \text{C mol of introduced substrate}) \times 100$$

Product yield (C-%)

$$= (\text{C mol of product} / \text{C mol of introduced substrate}) \times 100$$

product selectivity (C-%) = (product yield/conversions) $\times 100$

H-D exchange reactions between ethanol and ethanol- d_6 (Cambridge Isotope Laboratories) were carried out at 300°C under almost the same reaction conditions as the ethanol conversion. The reaction products were analyzed by GC-MS (Hewlett-Packard, HP-5972) with a DB-WAX capillary column (60 m, 0.25 mm i.d.).

3. Results and discussion

3.1. Hydrothermal synthesis of Sr-HAP with various Sr/P molar ratios

Fig. 1 shows XRD patterns of prepared strontium phosphate hydroxyapatites (Sr-HAP) powders with various Sr/P molar ratios, shown in parentheses in sample names. The bulk Sr/P molar ratios were determined by ICP analysis (Table 1). All patterns were attributable to Sr-HAP (PDF #33-1348) with a single phase. In addition, diffraction peak intensities and full widths at half maximum were also identical, indicating that all prepared samples had almost the same crystal size and crystallinity.

Fig. 2 shows FT-IR spectra of the prepared Sr-HAP with various Sr/P molar ratios. All samples showed an OH vibration band at

around 3590 cm^{-1} , indicating the existence of the OH groups found in apatite structures.

Table 1 shows the chemical compositions and specific surface areas of the prepared Sr-HAP samples. Although Sr/P molar ratios in raw solutions were 1.50, 1.60, 1.67, and 2.0, bulk Sr/P molar ratios of the prepared samples were determined by ICP to be 1.58, 1.64, 1.67, and 1.70. The bulk Sr/P molar ratios increased with the increase of the Sr/P molar ratio in raw solutions. The bulk Sr/P ratios were closed toward 1.67 from their Sr/P ratios in raw solutions. The ICP measurement error for Sr/P molar ratio was ± 0.01 . Sodium ions were detected in the samples with low Sr/P molar ratios (≤ 1.67), and the Na^+ ion content increased with the decreasing Sr/P molar ratio. These results implied the replacement of Sr^{2+} ions by Na^+ ions in samples with low Sr/P molar ratios. Although the Sr-HAP (1.70) sample showed a slightly higher specific surface area of $37.6 \text{ m}^2 \text{ g}^{-1}$, the other samples showed almost the same specific surface areas at $29.4\text{--}30.8 \text{ m}^2 \text{ g}^{-1}$.

Table 2 shows XPS data for the samples. The binding energies of all elements did not depend on the bulk Sr/P molar ratio and were in good agreement with Refs. [23–25]. However, the near-surface Sr/P molar ratio increased and the near-surface Na/P molar ratio decreased with the increase in the bulk Sr/P molar ratio. These results suggest that the near-surface composition is closely related to the bulk composition. The near-surface Sr/P atomic ratios of the samples were between 1.84 and 2.02, slightly higher than the samples' bulk Sr/P molar ratios, which might be due to the fact that strontium rich faces, such as (100), were preferentially exposed on the surface.

3.2. Catalytic conversion of ethanol over Sr-HAP catalysts

Catalytic conversion of ethanol was carried out at 300°C over the prepared Sr-HAP catalysts. Table 3 shows the catalytic activities and product selectivities. In the absence of the catalyst, ethanol conversion was nearly 0%, which indicated that the homogeneous reaction contributed not at all to the catalytic reaction. The specific reaction rates depended markedly on the catalysts' Sr/P molar ratios and increased as the Sr/P ratios increased. 1-Butanol selectivity also increased with increasing Sr/P molar ratio of the catalysts. With regard to by-products, selectivity for 2-buten-1-ol decreased but selectivity for C_6 and C_8 alcohols increased with the increase in Sr/P molar ratio from 1.58 to 1.67. Although the reaction rates were gradually decreased in the initial period over all Sr-HAP catalysts

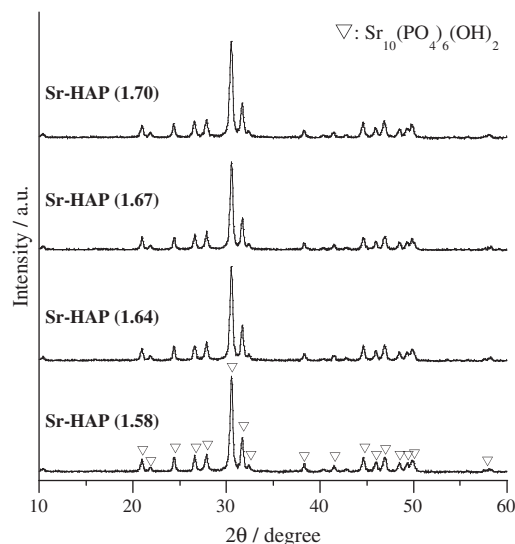
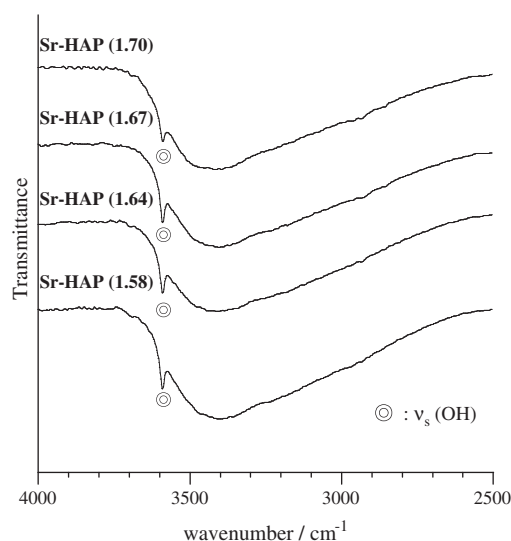


Table 1
Characteristics of Sr-HAP with various Sr/P molar ratios.

Sample	Sr/P molar ratio in raw solution	Relative atomic ratios			Sr/P molar ratio in solid product	Specific surface area (m ² g ⁻¹)
		Sr	P	Na		
Sr-HAP (1.58)	1.50	9.5	6.0	0.50	1.58	30.8
Sr-HAP (1.64)	1.60	9.8	6.0	0.28	1.64	29.6
Sr-HAP (1.67)	1.67	10.0	6.0	0.04	1.67	29.4
Sr-HAP (1.70)	2.00	10.2	6.0	n.d.	1.70	37.6

**Fig. 2.** FT-IR spectra of Sr-HAP with various Sr/P molar ratios.**Table 2**
Results of XPS analysis of Sr-HAP with various Sr/P molar ratios.

Sample	Binding energy (eV)			Molar ratios	
	Sr3p _{3/2}	P2s	Na1s	Sr/P	Na/P
Sr-HAP (1.58)	269.3	190.6	1072.4	1.84	0.35
Sr-HAP (1.64)	269.4	190.8	1072.8	1.90	0.11
Sr-HAP (1.67)	269.4	190.6	1072.3	1.96	0.06
Sr-HAP (1.70)	269.3	190.7	1072.4	2.02	0.03

as well as over the reported catalysts [26], the initial reaction rates were completely recovered by calcination treatments at 550 °C for 3 h in Ar flow (data not shown).

As shown in Fig. 3, 1-butanol selectivity depended not only on the catalyst's Sr/P molar ratio but also on the ethanol conversion. All Sr-HAP catalysts exhibited the highest 1-butanol selectivity in the region of ethanol conversion between 7% and 9%. In the case

of low ethanol conversion, below 7%, main by-products were intermediates such as 2-buten-1-ol and aldehydes, whereas in the case of high ethanol conversion, over 9%, main by-products were successive products, such as C₆ and C₈ alcohols, similarly to previous reports [12,15]. On the other hand, 1-butanol selectivity increased with the increase in the catalysts' Sr/P molar ratio, whose order was independent of the ethanol conversion rate.

Comparing the product selectivities in the region of ethanol conversion between 7% and 9% (Table S1), Sr-HAP (1.58) and Sr-HAP (1.64) catalysts exhibited higher selectivity for 2-buten-1-ol and lower selectivity for 1-butanol than Sr-HAP (1.67) catalyst. On the other hand, Sr-HAP (1.70) catalyst exhibited lower selectivity for C₆ and C₈ alcohols than Sr-HAP (1.67) catalyst. These results suggest that Sr-HAP catalysts with a high Sr/P molar ratio may accelerate the hydrogenation of intermediates, such as 2-buten-1-ol, into 1-butanol but also inhibit the successive reaction of C₄ products into C₆ and C₈ alcohols.

We postulated that the changes in product selectivity as the catalysts' Sr/P molar ratio changed might be due to a difference in the hydrogenation activities of the Sr-HAP catalysts. C₆ and C₈ alcohols may be formed via the aldol condensation of butyraldehyde produced by the hydrogenation of crotonaldehyde or the dehydrogenation of 1-butanol. The Sr-HAP (1.70) catalyst with high 1-butanol selectivity might preferentially accelerate the hydrogenation of butyraldehyde species for aldol condensation.

3.3. Rate-determining step of 1-butanol formation from ethanol

The reaction mechanism of 1-butanol formation from ethanol, as described in Scheme 1 [14–20], has previously been discussed. In this paper, we examined the rate-determining step in the reaction mechanism over a Sr-HAP catalyst with a Sr/P molar ratio of 1.67. When the initial ethanol concentration was varied between 10.3 and 53.5 mol%, 1-butanol selectivity was almost constant at around 81 C-% and the specific reaction rates decreased significantly from 24 to 17 μmol h⁻¹ m⁻² (Table S2). The reaction order was -0.13 in the ethanol concentration, which indicated that the adsorption of ethanol was not the rate-determining step in the formation of 1-butanol.

Table 3
Catalytic conversion of ethanol over Sr-HAP catalysts with various Sr/P molar ratios.^a

Catalyst	Conv. (C-%)	Selectivity (C-%)				Specific reaction rate (μmol h ⁻¹ m ⁻²)
		Acetaldehyde	1-Butanol	2-Buten-1-ol	C ₆ ,C ₈ -OH	
No catalyst	Trace	n.d. ^b	n.d. ^b	n.d. ^b	n.d. ^b	–
Sr-HAP (1.58)	1.1	1.5	69.0	22.3	2.6	2.7
Sr-HAP (1.64)	5.9	0.7	78.1	6.5	10.7	15.5
Sr-HAP (1.67)	7.9	0.5	81.7	3.0	12.0	20.6
Sr-HAP (1.70)	11.3	0.7	86.4	3.0	7.8	23.3
Sr-HAP-NH ₃ (1.71) ^c	4.4	0.4	79.2	5.8	8.8	8.4

^a Catalyst 2.0 g, temperature 300 °C, W/F_{ethanol} 130 h g mol⁻¹, ethanol 16.1 mol% (Ar balanced), time on stream 3 h.

^b Not detected.

^c This catalyst was prepared by almost the same method as the Sr-HAP (1.70) except for using NH₃ instead of NaOH as the alkaline source. The Sr/P molar ratio and specific surface area were 1.71 and 40.0 m² g⁻¹.

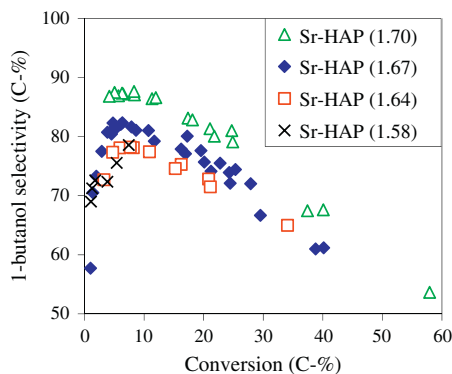


Fig. 3. Changes in 1-butanol selectivity with increasing ethanol conversion over Sr-HAP catalysts with various Sr/P molar ratios.

The catalytic conversion of 1-butanol was carried out over Sr-HAP catalyst (Table S3). Although 1-butanol showed a slightly lower specific reaction rate and a slightly higher selectivity for dimer alcohols such as C_8 alcohols than ethanol, the reaction behavior of 1-butanol was similar to that of ethanol. These results suggest that during ethanol conversion, the yield of successive reaction products increases with the increasing yield of 1-butanol and that 1-butanol desorption is probably not the rate-determining step in the formation of 1-butanol from ethanol.

Next, the catalytic conversion of acetaldehyde and crotonaldehyde, which are considered intermediates in the formation of 1-butanol from ethanol [14–20], was carried out in the presence of 2-butanol (Table S4). 2-Butanol was added as a hydrogen source, because we found that 1-butanol was not formed from 2-butanol and hydrogenation using hydrogen gas did not proceed over the Sr-HAP (1.67) catalyst (data not shown). In the catalytic conversion of crotonaldehyde, 1-butanol was formed with high yield, while C_2 compounds such as ethanol and acetaldehyde were not formed. These results indicated that the rate-determining step was prior to the hydrogenation of crotonaldehyde in the reaction mechanism. On the other hand, in the catalytic conversion of acetaldehyde, the ethanol yield was about 90% and the 1-butanol yield was low, indicating that the hydrogenation of acetaldehyde was much faster than the aldol condensation of acetaldehyde. In H–D exchange reactions carried out between H-ethanol and D-ethanol at 300 °C (Table S5 and Fig. S1), the H–D exchange reaction between ethanol molecules was much faster than ethanol conversion into 1-butanol, indicating that the dehydrogenation of ethanol was not the rate-determining step.

These results suggested that the rate-determining step in the formation of 1-butanol from ethanol is after acetaldehyde formation and before crotonaldehyde hydrogenation in Scheme 1—in other words, the aldol condensation of acetaldehyde. We postulated that the aldol condensation process involving the formation of two acetaldehyde adsorbates on neighboring basic sites just before condensation might take place relatively slowly on the Sr-HAP catalyst at 300 °C.

3.4. Characterization of acidic and basic properties

The TPD of CO_2 and NH_3 was measured to evaluate the acidic and basic properties of Sr-HAP catalysts with various Sr/P molar ratios. Fig. 4 shows the TPD profiles of CO_2 adsorbed onto Sr-HAP catalysts. All Sr-HAP catalysts had two types of basic sites. The relatively strong basic sites showed CO_2 desorption at around 200 °C, and the relatively weak basic sites showed CO_2 desorption at around 100 °C. The CO_2 -TPD intensity at around 200 °C increased as the catalyst's Sr/P molar ratio increased, indicating that the

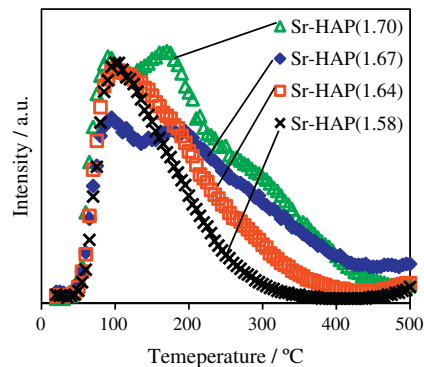


Fig. 4. TPD profiles of CO_2 adsorbed onto Sr-HAP catalysts with various Sr/P molar ratios.

relatively strong basic sites increased with the increase in Sr/P molar ratio.

Fig. 5 shows the TPD profiles of NH_3 adsorbed onto Sr-HAP catalysts. All Sr-HAP catalysts had relatively weak acidic sites that showed NH_3 desorption around 100 °C. All samples except the Sr-HAP (1.58) catalyst also had relatively strong acidic sites, which resulted in NH_3 desorption at over 200 °C (Fig. 5b). The NH_3 -TPD intensity at around 250 °C increased in the order Sr-HAP (1.58) < Sr-HAP (1.64) = Sr-HAP (1.67) < Sr-HAP (1.70), indicating that the relatively strong acidic sites increased with the increase in the Sr/P molar ratio. These results could be explained by the Lewis acidity of Na^+ ions, which was weaker than that of Sr^{2+} ions [27].

The FT-IR spectrum of pyridine adsorbed onto the Sr-HAP (1.58), which was a strontium-deficient apatite, is shown in Fig. S2. The peaks in the spectrum corresponded with hydrogen-bonded pyridine and coordinately bonded pyridine on Lewis acidic site at 1615, 1590, 1575, 1488, and 1440 cm^{-1} [28]. On the other hand, no peaks were observed at around 1540 cm^{-1} , which was assignable to pyridinium ions adsorbed onto Brønsted acidic sites [28]. These results suggested that the Sr-HAP has no Brønsted acidic sites.

The relatively strong acidic and basic site densities of Sr-HAP catalysts were estimated by measuring the adsorbed amounts of NH_3 and CO_2 at 250 °C, as shown in Table 4. All samples except the Sr-HAP (1.58) catalyst had these sites, and the acidic site density increased with the increase in Sr/P molar ratios, which corresponded with the NH_3 -TPD data shown in Fig. 5. The basic site density of the samples was significantly higher than the acidic site density. The basic site density increased from 0.37 to 1.00 $\mu mol\ m^{-2}$ as the Sr/P molar ratio increased. Thus, Sr-HAP catalysts with higher Sr/P ratios had higher densities of relatively strong basic sites, which also corresponded with the CO_2 -TPD data shown in Fig. 4.

3.5. Catalytic active sites

Tables 3 and 4 and Fig. 4 suggest a strong association between the catalytic activity of the Sr-HAP catalysts and the density of the relatively strong basic sites. As mentioned above, we speculate that the rate-determining step of 1-butanol formation is aldol condensation, which requires neighboring basic sites as the active sites on the catalysts. Therefore, Sr-HAP catalysts with higher Sr/P molar ratios, which show a corresponding higher density of relatively strong basic sites, could be expected to show higher catalytic activity for the formation of 1-butanol from ethanol.

From these results and data on ionic radii and electronegativities [29], we speculate that the relatively strong acidic sites on

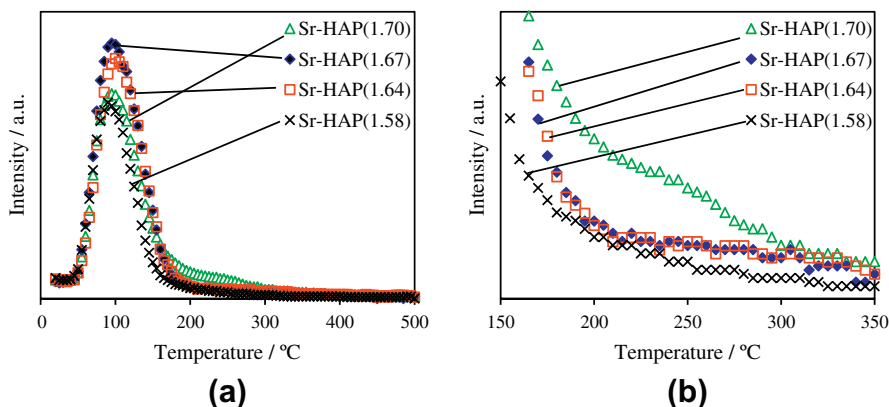


Fig. 5. (a) TPD profiles of NH_3 adsorbed onto Sr-HAP catalysts with various Sr/P molar ratios and (b) the same expanded.

Table 4

Acidic and basic site density of Sr-HAP catalysts with various Sr/P molar ratios.

Sample	Acidic site density ($\mu\text{mol m}^{-2}$) ^a	Basic site density ($\mu\text{mol m}^{-2}$) ^b
Sr-HAP (1.58)	<0.01	0.37
Sr-HAP (1.64)	0.13	0.58
Sr-HAP (1.67)	0.11	0.79
Sr-HAP (1.70)	0.22	1.00

^a Estimated from the amount of NH_3 adsorption at 250 °C.

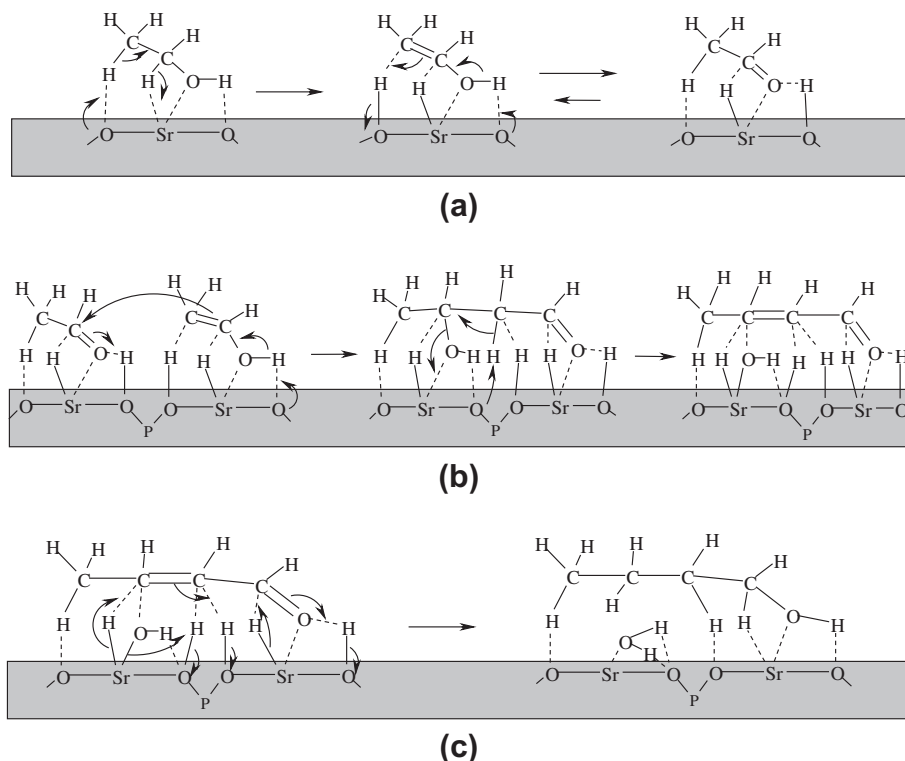
^b Estimated from the amount of CO_2 adsorption at 250 °C.

the Sr-HAP catalysts are strontium cations and the relatively strong basic sites are oxygen anions adjacent to strontium ions. We propose the reaction mechanism for 1-butanol formation from ethanol over Sr-HAP catalyst shown in Scheme 2. In this scheme, ethanol is hydrogen-dissociatively adsorbed as aldehyde or enol species (a).

The ethanol-derived adsorbates and hydride-like hydrogen are located on Lewis acidic sites comprising strontium atoms, and proton-like hydrogen is located on basic sites comprising oxygen atoms. Next, the aldehyde species react with the neighboring enol species to form unsaturated C_4 aldehyde species (aldol condensation) (b). Finally, 1-butanol is formed via hydrogenation of the unsaturated C_4 aldehyde species using the hydride-like and proton-like hydrogen (c).

3.6. Effect of composition on the catalytic properties

Based on the proposed structure of catalytic active sites described in Scheme 2, we postulate that the effects of the Sr/P molar ratio of the Sr-HAP catalysts on catalytic activity and selectivity are as follows. Sr-HAP catalysts with lower Sr/P ratios (<1.67) have a smaller number of relatively strong basic sites, such as O–Sr–O sites, to catalyze dehydrogenation and aldol condensation, because Sr^{2+} ions have been replaced by Na^+ ions in the catalysts. On the



Scheme 2. The reaction mechanism of 1-butanol formation from ethanol over Sr-HAP catalysts: (a) dehydrogenation and keto–enol tautomerism; (b) aldol condensation; (c) hydrogenations.

other hand, Sr-HAP catalysts with higher Sr/P ratios (>1.67) have a larger number of relatively strong basic sites. Although the XRD pattern of Sr-HAP (1.70) showed a single apatite phase, the highly dispersed SrO species might be formed on the surface [3] and might act as relatively strong basic active sites, together with the catalysts' active sites. However, a SrO/SiO₂ catalyst showed low catalytic activity and 1-butanol selectivity (Table S6). These results suggested that SrO species alone have too little activity, but dispersed SrO species on Sr-HAP might play a significant role in 1-butanol formation from ethanol. Moreover, surface hydrogen species would increase with the increase in dehydrogenated adsorbates, such as aldehyde and enol species, and this would lead to acceleration of the hydrogenation rate of aldehyde species and suppression of the formation of unsaturated by-products and successive reaction products. The fact that Sr-HAP catalysts with higher Sr/P ratios exhibited higher catalytic activity and selectivity for 1-butanol could be ascribed to their catalytic properties for the formation of ethanol dehydrogenated adsorbates and hydrogen adsorbates.

To clarify the effect of Na⁺ ions on the catalytic properties of Sr-HAP catalysts, we also prepared Sr-HAP-NH₃ (1.57) and Sr-HAP-NH₃ (1.71) catalysts without Na⁺ ions. The Sr-HAP-NH₃ (1.57) and Sr-HAP-NH₃ (1.71) showed almost the same Sr/P molar ratios (1.57 and 1.71) and XRD pattern as those of Sr-HAP (1.58) and Sr-HAP (1.70), respectively. As shown in Table 3 and Table S6, the Sr-HAP-NH₃ (1.57) showed catalytic activity similar to but selectivity different from those of the Sr-HAP (1.58). The Sr-HAP-NH₃ (1.57) catalyst had relatively high selectivities toward ethylene and diethyl ether, which were due to acid catalysis, whereas ethylene and diethyl ether did not form over the Sr-HAP (1.58) catalyst. In the case of Sr-HAP-NH₃ (1.57), the Sr²⁺ ions deficiency might be compensated by H⁺ ions, and resulted in the formation of acidic sites to produce ethylene and diethyl ether from ethanol. In contrast, in the presence of Na⁺ ions in the synthetic solution of Sr-HAP catalysts with Sr/P molar ratios lower than 1.67, Na⁺ ions were incorporated into the Sr²⁺ deficiency apatite structure, and the incorporated Na⁺ ions might prevent the formation of Brønsted acidic sites. These results are supported by NH₃-TPD (Figs. 5 and S3) and ethanol conversion studies of Sr-HAP and Sr-HAP-NH₃ (Tables 3 and S6).

On the other hand, although Sr-HAP (1.70) would contain slightly greater amounts of sodium ions than Sr-HAP-NH₃ (1.71), both catalysts had similar compositions and surface areas. However, Sr-HAP (1.70) catalysts showed activity and selectivity significantly higher than those of Sr-HAP-NH₃ (1.71) catalysts in ethanol conversion. As shown in TEM images of Fig. S4, the average aspect ratio of Sr-HAP-NH₃ (1.71) particles was obviously higher than that of Sr-HAP (1.70) particles. The difference in catalytic activities might be due to exposed surfaces depending on the crystallite morphologies, which might be affected by the pH of the synthetic solutions containing NaOH or ammonia. Based on this, although NaOH affected the catalytic properties for ethanol conversion, it was considered that the incorporated Na⁺ ions would not act as catalytic active sites.

4. Conclusions

Sr-HAP catalysts with Sr/P molar ratios between 1.58 and 1.70 were synthesized under hydrothermal conditions. The prepared Sr-HAP catalysts with higher Sr/P molar ratios showed higher catalytic activity and 1-butanol selectivity in the catalytic conversion of ethanol at 300 °C. XPS analysis revealed that near-surface Sr/P atomic ratios of the catalysts showed good correlations with their

bulk Sr/P molar ratios. Both relatively strong acidic and basic site densities of the Sr-HAP catalysts increased with the increase in the Sr/P molar ratio, although the basic site density was significantly higher than the acidic site density. The rate-determining step in 1-butanol formation from ethanol is postulated to be aldol condensation, including the condensation process itself and the formation of two aldehyde adsorbates on neighboring basic sites just before condensation. Catalysts with higher Sr/P molar ratios had higher densities of relatively strong basic sites, which could explain why Sr-HAP catalysts with higher Sr/P molar ratios show higher catalytic activity and 1-butanol selectivity in the catalytic conversion of ethanol.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jcat.2012.08.019>.

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